

Smart Electricity Management System

Using Data Structures and Algorithms

Ruhani Aneja Suryansh Tayal Aditi Gupta

Graphic Era (Deemed to be University), Dehradun
Department of Computer Science & Engineering
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Team ID: DSCPP-III-2026-T127
Mentor: Ms. Shrishti Chamoli

Team & Mentor Details

Team Information

- **Team Name:** Tech Titans
- **Team ID:** DSCPP-III-2026-T127
- **Semester:** 3rd
- **Domain:** Smart Energy Management

Team Members

- **Ruhani Aneja** – 2510370225 – Team Lead – **Section: ML3**
- **Suryansh Tayal** – 2510370336 – **Section: ML4**
- **Aditi Gupta** – 2510380082 – **Section: DS2**

Mentor: Ms. Shrishti Chamoli

Interactions Completed: 3

Problem Statement & Motivation

Problem Statement

Users often know their electricity bill but do not know which appliances consume more energy or whether they are being used during expensive periods.

The project aims to provide a simple system that analyses appliance usage and gives useful suggestions for reducing energy consumption and cost.

Motivation

- High electricity consumption can increase monthly bills.
- Users may not notice the effect of peak-hour usage.
- Basic monitoring mainly shows consumption instead of suggesting what the user can change.
- Appliance-wise analysis can turn usage data into practical energy-saving suggestions.

Target Users / Stakeholders

Project Objectives

Key Objective

Develop a C++ based Smart Electricity Management System that analyses appliance usage, estimates energy cost and suggests better usage schedules.

- Maintain a predefined database of common electrical appliances.
- Take simple user input: **Appliance Name + Start Time + End Time.**
- Calculate appliance-wise energy consumption.
- Estimate electricity cost using predefined tariff data.
- Identify peak and off-peak usage.
- Generate alerts for high-cost usage periods.
- Suggest better operating times for schedulable appliances.
- Provide energy-saving and What-If recommendations.
- Generate energy reports and budget alerts.
- Apply Data Structures and Algorithms to a practical real-world problem.

Proposed Solution

Input / Data Collection

The user provides only:

- Appliance Name
- Start Time
- End Time

Other appliance details are automatically retrieved from the predefined appliance database.

Processing / Core Algorithm

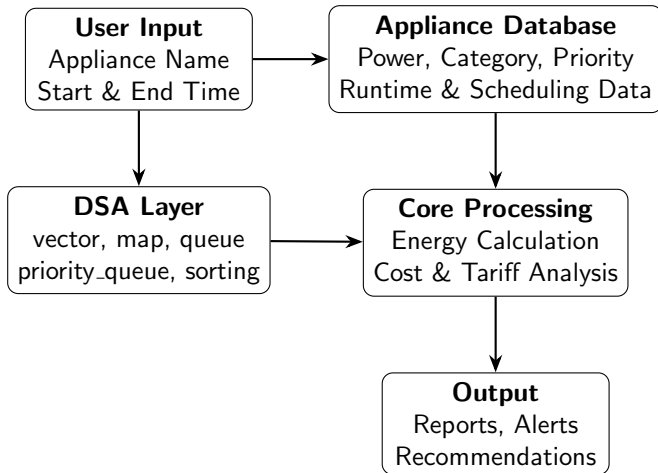
- Retrieve the appliance profile.
- Calculate usage duration and energy consumption.
- Check peak / off-peak period.
- Calculate estimated electricity cost.
- Analyse scheduling possibilities.

Integration of PBL Course Concepts

Course / Area	Concepts Applied	Project Module
Data Structures in C	Arrays, searching, sorting, queues, priority queues and data organisation	Appliance records, usage analysis and scheduling
OOPs with C++	Classes, objects, encapsulation, inheritance, polymorphism and abstraction	Appliance management and different appliance behaviours
STL	vector, map, queue, priority_queue and set	Appliance storage, tariff lookup, pending requests and scheduling
File Handling	File input/output and persistent records	Usage history and energy reports

Meaningful Integration

Data Structures and Object-Oriented Programming are used together in the same system to manage, analyse and process electricity usage data.



Technology Stack & Methodology

Technology Stack

- **Programming:** C++
- **Paradigm:** Object-Oriented Programming
- **DSA:** STL and core data structures
- **Storage:** File-based Storage
- **IDE:** Visual Studio Code
- **Version Control:** Not Used

Development Methodology

- ① Requirement Analysis
- ② System Design
- ③ Appliance Database Design
- ④ C++ Implementation
- ⑤ DSA Integration
- ⑥ Testing
- ⑦ Evaluation

Technology Selection

C++ is suitable because it allows us to combine Object-Oriented Programming with STL data structures and algorithms. File-based storage keeps the project simple while allowing usage records and reports to be maintained.

Initial Progress & Team Contribution

Progress Achieved

- Background and literature study completed.
- Project requirements identified.
- Major system features defined.
- Predefined appliance database prepared.
- Appliance profile structure designed.
- DSA and C++ concepts mapped to project modules.
- High-level system architecture prepared.
- Initial project planning and documentation completed.

Member	Role	Contribution
Ruhani Aneja	Team Lead	Project planning, system design, documentation and team coordination
Suryansh Tayal	Developer	DSA planning, appliance management and core system

Roadmap, Expected Outcomes & References

Roadmap

- **Phase I:** Proposal & Design
- **Phase II:** Development
- **Phase III:** Final Implementation

Expected Outcomes

- Working C++ prototype
- Appliance-wise energy analysis
- Cost and tariff analysis
- Smart scheduling
- Energy-saving recommendations
- Reports and alerts

References

- ① Uttarakhand Electricity Regulatory Commission (UERC) – Tariff Orders.
- ② UPCL – Electricity Tariff Documentation.
- ③ C++ Standard Library Documentation.
- ④ Reference material on Data Structures and Algorithms using C++.

Thank You!

Questions & Suggestions